

CS5275 – The Algorithm Designer's Toolkit
(S2 AY2025/26)

Lecture 14:

Boolean functions – learning theory

Computational learning theory

Input. We are given a set of training examples:

$$(x, f(x)),$$

where f is an unknown function.

Goal. Learn a hypothesis function h that is good at predicting $f(y)$ for future input y .

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This captures many real-life applications:

- Movie recommendation.
- Spam email detection.
- Credit card fraud detection.
- Weather prediction.
- ...

PAC learning

- The model of **P**robably **A**pproximately **C**orrect learning:

A **concept class** $\mathcal{C}$ is a collection of functions $\{-1,1\}^n \rightarrow \{-1,1\}$.

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There are two ways to access an unknown function $f \in \mathcal{C}$.

- **Random examples:** Draw $(\mathbf{x}, f(\mathbf{x}))$, where $\mathbf{x} \sim \{-1,1\}^n$.
- **Queries:** Request the value $f(x)$ for any $x \in \{-1,1\}^n$.

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Given two parameters (ϵ, δ) , the goal is to compute a hypothesis $h : \{-1,1\}^n \rightarrow \{-1,1\}$ such that $\Pr[\text{dist}(f, h) \leq \epsilon] \geq 1 - \delta$.

PAC learning

A framework that provides a theoretical guarantee for learning:

- How much data is enough?
- Can a function be learned at all?

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- **Random examples:** Draw $(\mathbf{x}, f(\mathbf{x}))$, where $\mathbf{x} \sim \{-1,1\}^n$.
- **Queries:** Request the value $f(x)$ for any $x \in \{-1,1\}^n$.

- Data is limited.
- True function f is unknown.

Given two parameters (ϵ, δ) , the goal is to compute a hypothesis $h : \{-1,1\}^n \rightarrow \{-1,1\}$ such that $\Pr[\text{dist}(f, h) \leq \epsilon] \geq 1 - \delta$.

- ϵ : How accurate do we want to be?
- δ : How confident do we want to be?

Spectral concentration

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- Let $\mathcal{F}$ be a collection of subsets $S \subseteq [n]$.
- The Fourier spectrum of $f : \{-1,1\}^n \rightarrow \mathbb{R}$ is **ϵ -concentrated** on $\mathcal{F}$ if

$$\sum_{S \subseteq [n], S \notin \mathcal{F}} \hat{f}(S)^2 \leq \epsilon.$$

Spectral concentration

Spectral concentration → efficient learning

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Theorem: Suppose we are given:

- a collection $\mathcal{F}$ on which the Fourier spectrum of $f : \{-1,1\}^n \rightarrow \{-1,1\}$ is $(\epsilon/2)$ -concentrated
- access to random examples of f

Then f can be learned in $\text{poly}\left(n, \frac{1}{\epsilon}, |\mathcal{F}|\right) \cdot O\left(\log \frac{1}{\delta}\right)$ time with accuracy ϵ and error probability δ .

Estimating one Fourier coefficient

Claim: $O\left(\frac{\log \frac{1}{\delta}}{\epsilon^2}\right)$ random examples of f suffice to obtain an estimate $\tilde{f}(S)$ of $\hat{f}(S)$ such that

$$\Pr[|\tilde{f}(S) - \hat{f}(S)| \leq \epsilon] \geq 1 - \delta.$$

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- Repeat for $O\left(\frac{\log\frac{1}{\delta}}{\epsilon^2}\right)$ times and take average:
- Sample $\mathbf{x} \sim \{-1, 1\}^n$ and calculate $f(\mathbf{x})\chi_S(\mathbf{x})$.

$$\hat{f}(S) = \langle f, \chi_S \rangle = \mathbb{E}[f(\mathbf{x})\chi_S(\mathbf{x})]$$

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Hoeffding inequality:

- $X_1, X_2, \dots, X_n$ are independent random variables taking values in $[a_i, b_i]$.
- $X = \sum_{i=1}^n X_i$.
- $\mu = \mathbb{E}[X]$.

$[-1, 1]$

$$\Pr[|X - \mu| \geq t] \leq 2e^{\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2}}$$

$$t = n \cdot \epsilon$$

$$2e^{\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2}} = 2e^{\frac{-\epsilon^2}{2n}} \leq \delta$$

$$n \geq \frac{2 \ln \frac{2}{\delta}}{\epsilon^2}$$

$f(\mathbf{x})\chi_S(\mathbf{x})$

$\mu = n \cdot \hat{f}(S)$

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Learning algorithm

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- For each $S \in \mathcal{F}$, calculate an estimate $\tilde{f}(S)$ such that

$$\Pr\left[|\tilde{f}(S) - \hat{f}(S)| \leq \sqrt{\frac{\epsilon}{2|\mathcal{F}|}}\right] \geq 1 - \frac{\delta}{|\mathcal{F}|}.$$

- $g = \sum_{S \in \mathcal{F}} \tilde{f}(S) \chi_S$

Output: • $h = \text{sgn}(g)$

$\text{sgn}(\cdot)$ ensures that the h is a function $\{-1, 1\}^n \rightarrow \{-1, 1\}$.

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With probability $1 - \delta$, for all $S \in \mathcal{F}$,

$$(\tilde{f}(S) - \hat{f}(S))^2 \leq \frac{\epsilon}{2|\mathcal{F}|}.$$

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Then f can be learned in $\text{poly}\left(n, \frac{1}{\epsilon}, |\mathcal{F}|\right) \cdot O\left(\log \frac{1}{\delta}\right)$ time with accuracy ϵ and error probability δ .

Analysis

$$\|g - f\|_2^2 = \sum_{S \subseteq [n]} (\widehat{g - f})(S)^2 = \underbrace{\sum_{S \in \mathcal{F}} (\widehat{g - f})(S)^2}_{\text{purple equation}} + \sum_{S \notin \mathcal{F}} (\widehat{g - f})(S)^2 \leq \epsilon$$

$$\sum_{S \in \mathcal{F}} (\tilde{f}(S) - \hat{f}(S))^2 \leq \sum_{S \in \mathcal{F}} \frac{\epsilon}{2|\mathcal{F}|} \leq \frac{\epsilon}{2}$$

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$$\sum_{S \notin \mathcal{F}} \hat{f}(S)^2 \leq \frac{\epsilon}{2}$$

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$$\text{dist}(f, h) = \Pr[f \neq h] \leq \mathbb{E}[(g - f)^2] = \|g - f\|_2^2 \leq \epsilon$$

If $f(x) \neq h(x)$, then $(g(x) - f(x))^2 \geq 1$

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Implications

Let $\mathcal{C}$ be any concept class such that:

- the Fourier spectrum of all $f : \{-1,1\}^n \rightarrow \{-1,1\}$ in $\mathcal{C}$ is $(\epsilon/2)$ -concentrated up to degree k .

$$\mathcal{F} = \{S \subseteq [n] : |S| \leq k\}$$

Low-degree algorithm

Then $\mathcal{C}$ can be learned from random examples in

$$\text{poly}\left(n^k, \frac{1}{\epsilon}\right) \cdot O\left(\log \frac{1}{\delta}\right) \text{ time}$$

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Implications

Small degree $\rightarrow$ efficient learning

Any bounded-degree function $f : \{-1,1\}^n \rightarrow \{-1,1\}$ can be learned from random examples in polynomial time.

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Let $\mathcal{C}$ be any concept class such that:

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Low-degree algorithm

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Claim: The Fourier spectrum of any $f : \{-1,1\}^n \rightarrow \mathbb{R}$ is ϵ -concentrated up to degree $\frac{I[f]}{\epsilon}$.

Implications

Small total influence $\rightarrow$ efficient learning

Any function $f : \{-1,1\}^n \rightarrow \{-1,1\}$
with $I[f] \leq t$ can be learned from
random examples in

$$n^{O(\frac{t}{\epsilon})} \cdot O\left(\log \frac{1}{\delta}\right) \text{ time.}$$



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Recall:

$$I[f] = \sum_{S \subseteq [n]} |S| \hat{f}(S)^2$$

$$I[f] > \sum_{|S| > \frac{I[f]}{\epsilon}} \frac{I[f]}{\epsilon} \cdot \hat{f}(S)^2$$

$$\epsilon > \sum_{|S| > \frac{I[f]}{\epsilon}} \hat{f}(S)^2$$

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$$\|v\|_1 \leq \sqrt{n} \cdot \|v\|_2 \qquad 1 = \sum_{S \subseteq [n]} \hat{f}(S)^2$$

$$I[f] = \sum_{i \in [n]} \hat{f}(\{i\})^2 \leq \sqrt{n \cdot \sum_{i \in [n]} \hat{f}(\{i\})^2} \leq \sqrt{n}$$

Recall: $f : \{-1,1\}^n \rightarrow \{-1,1\}$ is monotone $\rightarrow \text{Inf}_i[f] = \hat{f}(\{i\})^2$

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Any **monotone** $f : \{-1,1\}^n \rightarrow \{-1,1\}$
can be learned from random examples in

$$n^{O\left(\frac{\sqrt{n}}{\epsilon}\right)} \cdot O\left(\log \frac{1}{\delta}\right) \text{ time.}$$

$$\|v\|_1 \leq \sqrt{n} \cdot \|v\|_2 \qquad 1 = \sum_{S \subseteq [n]} \hat{f}(S)^2$$

$$I[f] = \sum_{i \in [n]} \hat{f}(\{i\})^2 \leq \sqrt{n \cdot \sum_{i \in [n]} \hat{f}(\{i\})^2} \leq \sqrt{n}$$

Recall: $f : \{-1,1\}^n \rightarrow \{-1,1\}$ is monotone $\rightarrow \text{Inf}_i[f] = \hat{f}(\{i\})^2$

Outlook

- **Next:** Suppose the Fourier spectrum of f is concentrated on a small collection of subsets. How do we find such a collection efficiently?



This is needed in order to apply the learning algorithm.

References

- Sections 3.1 and 3.4 of <https://arxiv.org/abs/2105.10386>